

AI3022: Search Methods in AI

Monsoon Semester 2025

Assignment 1 - Part 2

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Due Date: September 16, 2025, 23:59

Search Algorithm Implementation

For your chosen problem in Part 1:

- Implement the DEPTH FIRST SEARCH Algorithm.
- Implement the BREADTH FIRST SEARCH Algorithm.
- Define a heuristic function and implement the BEST FIRST SEARCH Algorithm.

Submit your code by the due date (*no extensions*). Along with the code, submit a small report describing your implementation and also performance of the three programs on the same problem, or a set of problems. Program demonstrations may be scheduled in due course.

› Submission Requirements

 This assignment must be submitted by a **single** team member, specifically the same individual who submitted Part 1.

- Create a **zip folder** with the following naming convention: ID1_ID2_ID3.zip where ID1, ID2, ID3 are the student ID numbers of all three team members.
- **Zip folder contents:**
 - code/ - Directory containing all source code files.
 - report.pdf - Implementation report describing your algorithms and performance analysis.